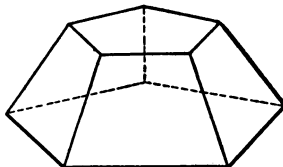


8. FRUSTUM OF REGULAR PYRAMID

Definition. A frustum of a regular pyramid is the portion of a regular pyramid included between the base and a section parallel to the base.



1. The slant height of a frustum of a regular pyramid is the altitude of a face.
2. The lateral edges of a frustum of a regular pyramid are equal, and the faces are regular equal trapezoids.

Notation

b = area of upper base.	p = perimeter of upper base.
B = area of lower base.	P = perimeter of lower base.
h = altitude.	S = lateral area.
l = slant height.	T = total area.

V = volume.

FORMULAS

The lateral area of the frustum of a regular pyramid is equal to one-half the sum of the perimeters of the bases multiplied by the slant height.

$$S = \frac{1}{2}(p + P)l.$$

The total area of the frustum of a regular pyramid is equal to the sum of the lateral area and the areas of the bases.

$$T = S + b + B = \frac{1}{2}(p + P)l + b + B.$$

The volume of the frustum of any pyramid is equal to one-third of the product of the altitude and the sum of the upper base, the lower base, and the mean proportional between the two bases.

$$V = \frac{h}{3}(b + B + \sqrt{bB}).$$